eResearch Australasia 2026 · Research Compute

Repurposing the ARDC BinderHub

On-demand web hosting on national research infrastructure

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LDaCA

THE UNIVERSITY OF SYDNEY
Sydney Informatics Hub

Australian Research Data Commons

The app

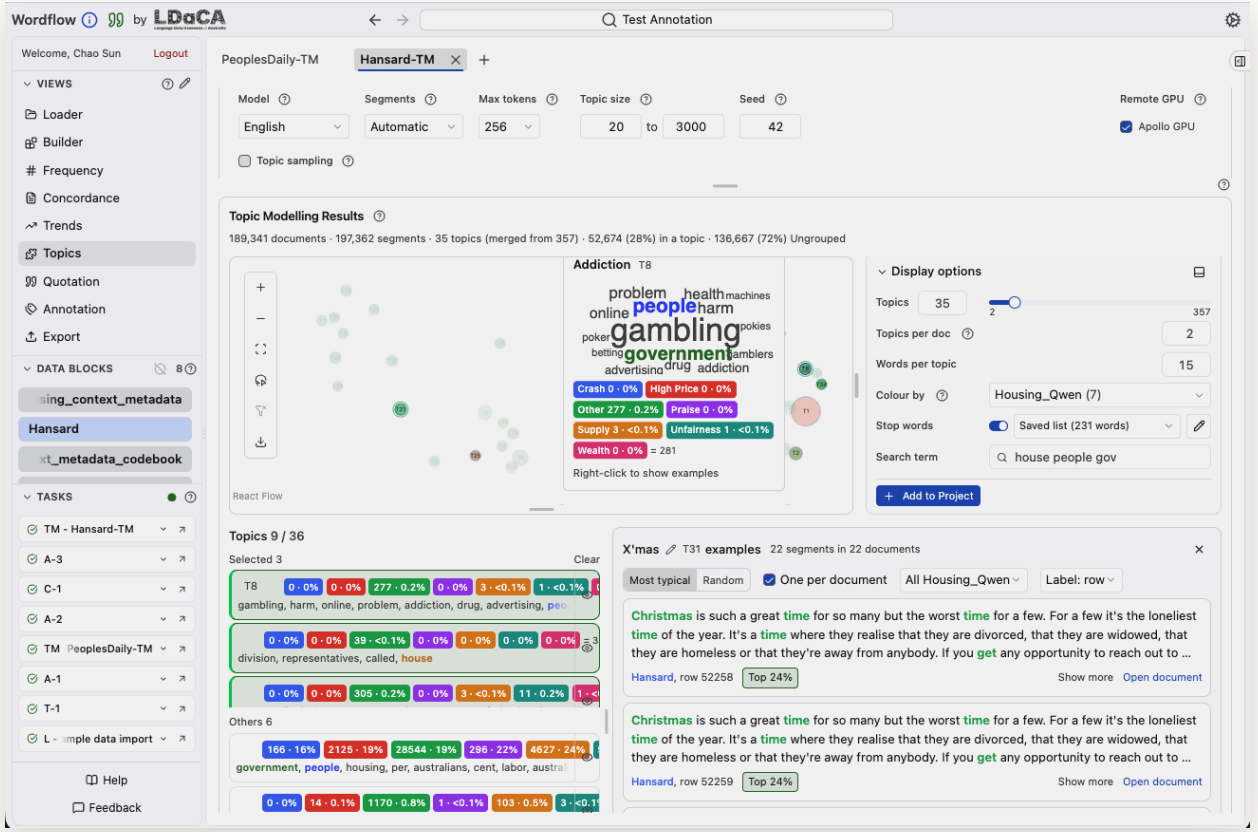
Wordflow: text analysis without programming

LDaCA's open-source, point-and-click app for researchers who work with language data: linguists, social scientists, educators, historians.

Frequency **Concordance** **Trends**

Topics **Quotation** **Annotation**

Load a corpus, run a tool, read the result beside the text. Most results can feed the next tool.



Open source · 39 releases since May 2026 · DOI 10.5281/zenodo.20408328

Repurposing the ARDC BinderHub · Chao Sun

How it runs

Two parts: a browser tab and a backend

Browser tab

The Wordflow interface. Nothing to install; any operating system.

Wordflow backend

Python with a fast data engine. Reads the files, counts, searches, models.

The Project

Files and results. Download it as one archive; open it anywhere Wordflow runs.

Nothing to install

Open a link, sign in, start.

Same app on every machine

Windows, macOS, a locked-down loan laptop.

Light jobs feel instant

Frequencies, searches and trends on a workshop-sized corpus.

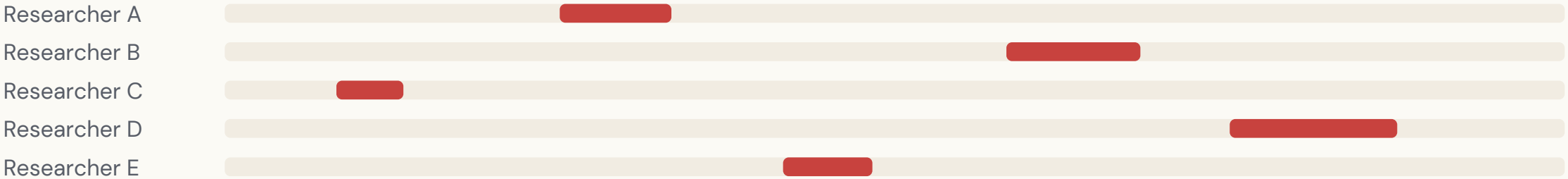
The interface is the same wherever the backend runs. **The question is where the backend lives.**

Scenario one: the individual researcher

A bursty need, a permanent server

Thinking, reading	Collecting data (e.g. interviews)	Preparing data	**Analysis**	Writing, verifying, revising
ongoing	months	weeks	days	months

For one researcher: a project of months or years; the tool matters for **the red part**, short and intense, and worked hard while it lasts.



For the provider: the same bursts, scattered. Sparse, unpredictable, never zero. A sketch, not measured data.

> **A standing server, even a small one, spends most of its life on standby, waiting for an ad-hoc user. Yet the tool must be there when that user arrives, and when a server is the only option, standby is the price.**

Scenario two: the training workshop

Thirty people at once, for three hours

A spike, on a known day

Everyone clicks at the same moment. The peak is the whole event.

Beyond our capacity

No budget for a server big enough for thirty backends at once.

Normal days between

Ad-hoc individual use on normal days; a peak-sized server stands by.



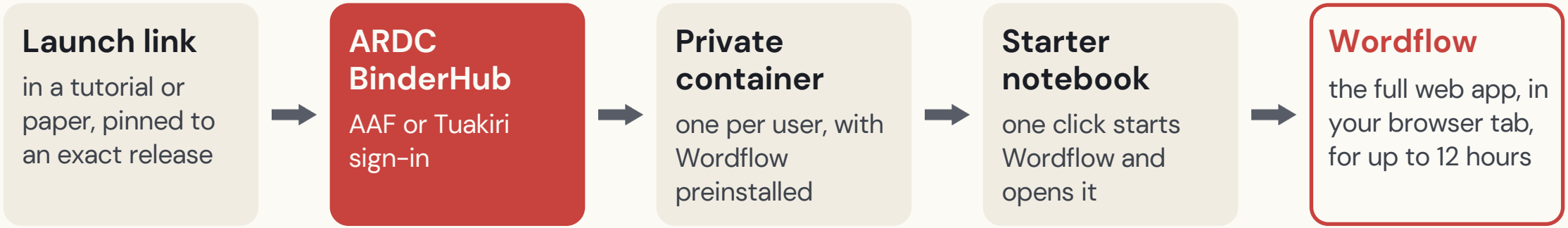
The shape of our workshop demand

Spikes on workshop days, the scattered individual use in between. A sketch, not measured data.

> **Two shapes of demand, one mismatch: a standing server is sized for a peak it meets a few days a year, and stands by for the rest.**

Hiding in plain sight: the ARDC BinderHub

Built to run notebooks. Underneath, an on-demand container launcher.



The image carries Wordflow; the notebook only starts it and hands you the interface.

To BinderHub, an ordinary notebook session. The ARDC BinderHub team supports this use.

Each container: 8 vCPU, 16 GB RAM, 16 GB temporary storage, up to 12 hours. Nothing is kept when the tab closes.

The whole trick

Two small files in a public repository

binder/environment.yml

```
dependencies:
  - python=3.14.5
  - pip:
      - ldaca-wordflow[deploy]==0.7.19
```

index.ipynb · one cell

```
server = await start_async_server(
    serve_frontend=True,
    port=8001,
    root_path=f"{PREFIX}/proxy/8001",
)
display(Markdown("# [Open Wordflow](...)"))
```

repo2docker builds the image
from the environment file, once per commit

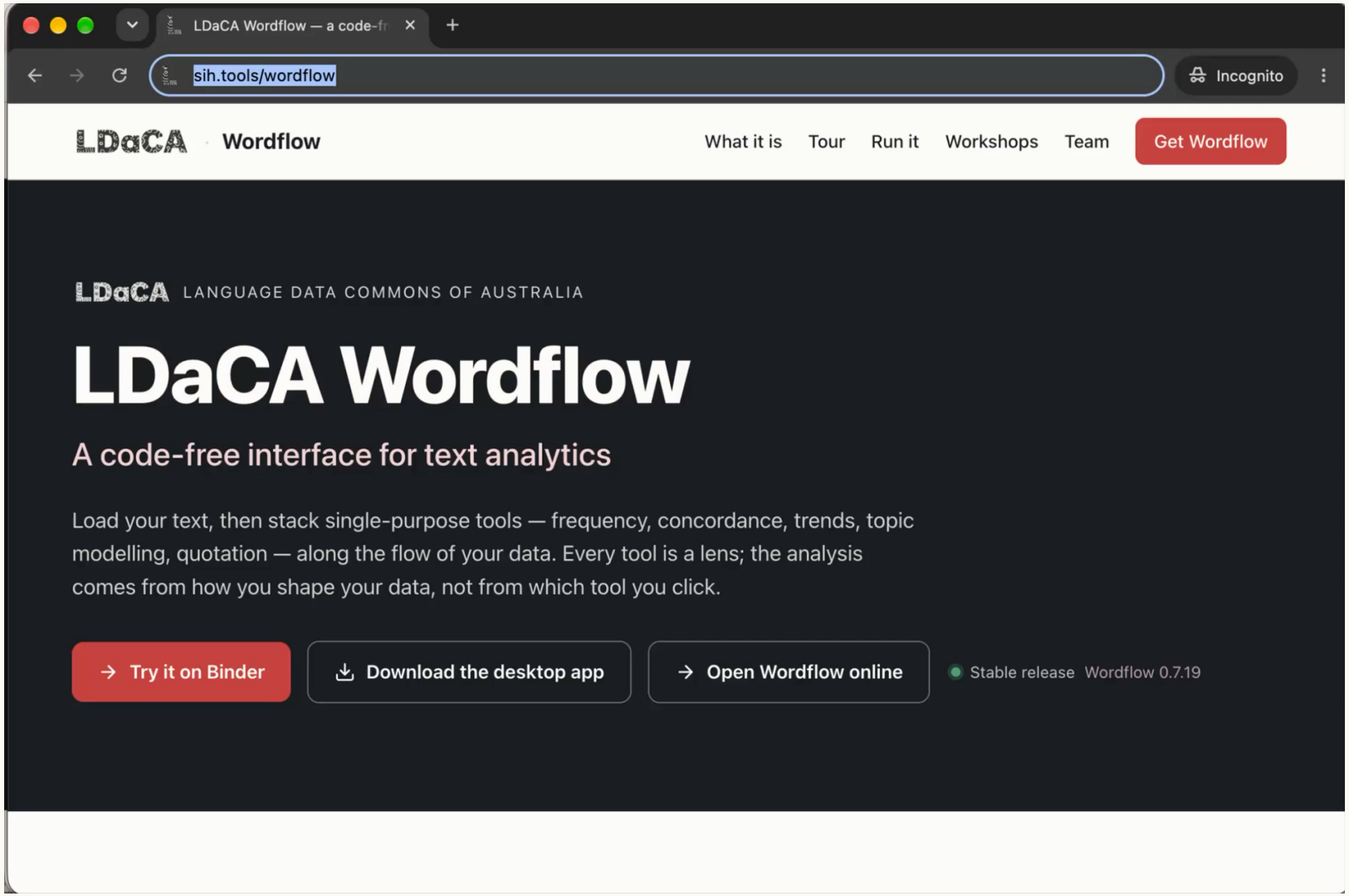
→

The notebook starts Wordflow
on a local port inside the container

→

Jupyter Server Proxy serves it
under the user's own signed-in URL

LDaCA



The mechanism, live

1. Landing page: Try it on Binder
2. Sign in with AAF
3. Container starts, notebook opens
4. One click: Wordflow

The launch link pins an exact release.

The same version opens every time, from a tutorial or a paper.

Click the video to play · 47 s · muted

Reproducible by construction

The launch link pins the release

```
binderhub.rc.nectar.org.au/v2/gh/Australian-Text-Analytics-Platform/
LDaCa_Text_Analytics_Tools/4b40d1e?labpath=index.ipynb
```

The commit
freezes the repository at one point in time

→

The repository
names one Wordflow version to install, here 0.7.19

→

The link
therefore opens that exact version, today or years from now

Every public release since May 2026 has its own launch link in the release history.
A workshop sheet or a methods section can name the exact version it used.

Could your tool, existing or planned, live on the ARDC BinderHub too?

Used like an app, or run as a service?

Used like an app: BinderHub can host it

- ✓ A person drives it, live
- ✓ One private copy per launch
- ✓ Work happens while you are there
- ✓ You bring data, you take results

Run as a service: it still needs a server

Runs unattended, in the background

Shared state, many users at once

Needs an always-on URL

Keeps data on the server

Wordflow is the left column. If your tool looks like that, or could be built to, a small repository of your own, set up the same way, is all it takes.

The weaknesses are the boundary

Sessions end

A service must stay up.

An app session just ends.

Cold starts

A service cannot keep users waiting.

An app user waits a minute.

No storage

A service keeps state.

An app user brings data, takes results, and downloads before closing the tab.

> **The three flaws that disqualify a service are the three things an app, and its user, can live with.**

What hosting Wordflow on the ARDC BinderHub buys us

In production since June 2026

0
servers to maintain for this channel

0
capacity worries before a workshop

1
pinned launch link for every release

5
workshops where Binder meant nothing to install

Sign-in comes with the platform: **AAF** for Australian and **Tuakiri** for New Zealand institutions. Wordflow's try-it-now channel since its first public release.

What it means

Existing infrastructure, serving a new kind of need

For researchers

Try it within minutes, with the login they already have, on a managed laptop where nothing can be installed.

For trainers

A full room on the same version within minutes. No setup to do beforehand, nothing to clean up afterwards.

For a small team

No system administration: no server to patch, secure or watch. Publishing a release is the whole deployment.

For the sector

The same investment, built for notebooks, now reaches researchers who do not code: point-and-click tools, on capacity it already has, inside the caps it already enforces.

Honest limits

A good tenant on shared infrastructure

Not the designed use

Yet every part behaves exactly as designed. We add nothing and bypass nothing.

No persistent storage

Work does not resume at the next login. Download your Project before you close the tab.

Per-user caps apply

8 vCPU, 16 GB RAM, sessions of up to 12 hours. An app-shaped tool fits inside them.

Australian and New Zealand logins only

Colleagues elsewhere use the desktop app instead.

We consulted the **ARDC BinderHub team** first; they support this use for research. We are grateful for the platform and for that conversation.

The pattern travels

A second tenant, in R: LADAL

Nine R Shiny text-analysis tools from the Language Technology and Data Analysis Laboratory, University of Queensland, now launch on the ARDC BinderHub.

Same approach: his own starter notebooks, the same sign-in, the same caps. Different language, different team.

Martin Schweinberger · LADAL is part of LDaCA · github.com/SLCLADAL/tools

- **WordFinder**
- **KeywordExtractor**
- **WordWebber**
- **POSTagger**
- **TextCleaner**
- **FileRenamer**
- **SentimentExplorer**
- **CollocationCalculator**
- **TopicDetector**

> **Nothing here is Wordflow-specific. It suits research tools with a browser interface that can run in a container, as ours and LADAL's do.**

Where we are

Our attempt at on-demand hosting, on infrastructure that already exists

We cannot promise it is the right way, but it has served our researchers since June. If a tool of yours is used like an app, we would like to compare notes.

Scan for the launch link, the repository and the desktop apps · DOI 10.5281/zenodo.20408328



Launch Wordflow on Binder

Thank you

To the **Australian Research Data Commons** and the **Nectar Research Cloud**, for BinderHub and for supporting this use.

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